FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – IX
PAPER –2
TEST DATE: 12-05-2024

ANSWERS, HINTS & SOLUTIONS

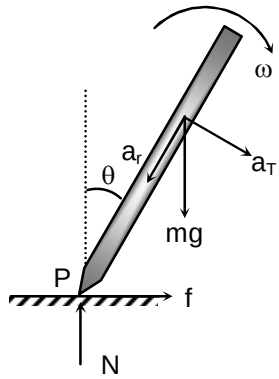
Physics

PART – I

SECTION – A

1. C

Sol. Using conservation of mechanical energy



$$mg\left(\frac{\ell}{2}-\frac{\ell}{2}\cos\theta\right)=\frac{1}{2}\left(\frac{m\ell^2}{3}\right)\omega^2$$

$$\omega^2=\frac{3g}{\ell}(1-\cos\theta) \quad \ldots(i)$$

Torque about point P.

$$mg\frac{\ell}{2}\sin\theta=\frac{m\ell^2}{3}\alpha$$

$$\alpha=\frac{3g}{2\ell}\sin\theta \quad \ldots(ii)$$

$$mg-N=m(a_r\cos\theta+a_T\sin\theta)$$

$$mg-N=m\left(\frac{\ell}{2}\omega^2\cos\theta+\frac{\ell}{2}\alpha\sin\theta\right) \quad \ldots(iii)$$

From (i), (ii) and (iii) we get

$$N=mg\left(\frac{3\cos\theta-1}{2}\right)^2$$

$$\frac{4}{9}mg=mg\left(\frac{3\cos\theta-1}{2}\right)^2$$

$$\theta=\cos^{-1}\left(\frac{7}{9}\right)$$

2. A

Sol. $\Delta x=d\sin\theta \Rightarrow 3\lambda=5\lambda\sin\theta \Rightarrow \sin\theta=\frac{3}{5}$

$$Y_n = D\tan\theta = \frac{3}{4}D$$

$$a_x = 2t \Rightarrow v_x = t^2 = \frac{dD}{dt} = 16$$

$$v_y = \frac{3}{4}\frac{dD}{dt} = \frac{3}{4}t^2 = 12$$

$$\vec{v} = v_x\hat{i} + v_y\hat{j}$$

$$v = \sqrt{v_x^2 + v_y^2} = 20 \text{ m/s}$$

3. B

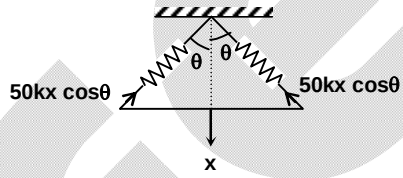
Sol. Net vertical force (as shown in figure)

$= 2\{(50kx\cos\theta)\cdot\cos\theta\} = 36kx = kx$

So, equivalent spring constant of the system in the given figure is = 36k

Similarly final equivalent spring constant is k/2

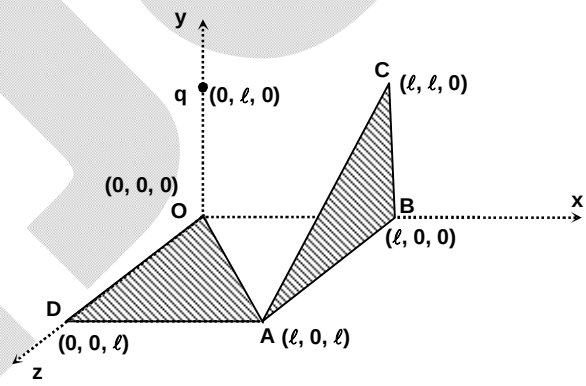
$$T = 2\pi\sqrt{\frac{m}{k/2}} = 2\pi\sqrt{\frac{2m}{k}}$$



4. C

Sol. $\phi = \phi_{AOD} + \phi_{ABC}$

$$= \frac{q}{48\varepsilon_0} + \frac{q}{48\varepsilon_0} = \frac{q}{24\varepsilon_0}$$



5. A, B, C, D

Sol. Energy stored in the space R to 2R

$$U = \int_R^{2R} \frac{1}{2}\varepsilon_0 E^2 dV = \frac{1}{2}\varepsilon_0\left(\frac{1}{4\pi\varepsilon_0}\cdot\frac{q}{r^2}\right)^2 \cdot 4\pi r^2.dr = \frac{kQ^2}{4R}$$

Similarly, energy stored in the space 4R to 6R is $\frac{kQ^2}{24R}$ and between 8R to ∞ is $\frac{kQ^2}{16R}$

6. A, C, D

Sol. $x = h \cot \theta$

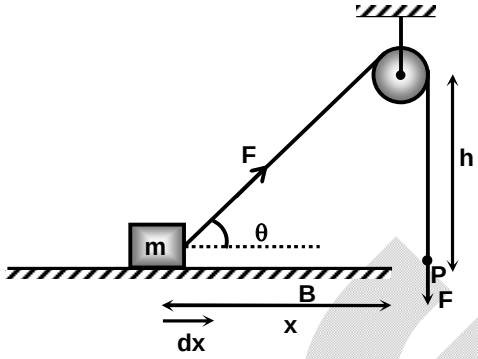
$dx = h \operatorname{cosec}^2 \theta \, d\theta$

$W = \int F\cos\theta dx$

$= \int F\cos\theta \cdot h \cdot \operatorname{cosec}^2\theta \cdot d\theta$

$W = Fh\left[-\frac{1}{\sin\theta}\right]_{\theta_1}^{\theta_2}$

$W_{A\to B} = -Fh\left\{\frac{1}{\sin 53^\circ} - \frac{1}{\sin 37^\circ}\right\} = \left(\frac{5}{12}Fh\right)$



Velocity of the block when angle is 53°

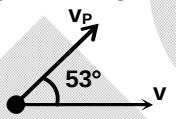
$\frac{1}{2}mv^2 = \frac{5}{12}Fh \Rightarrow v = \sqrt{\frac{5Fh}{6m}}$

Power = $P = \left(\sqrt{\frac{5Fh}{6m}}\right) \cdot F\cos 53^\circ = \sqrt{\frac{3F^3h}{10m}}$

Velocity of point P, when the angle made by the string becomes 53° then,

$v_P = v\cos 53^\circ$

$v_P = \sqrt{\frac{5Fh}{6m}} \cdot \frac{3}{5} = \sqrt{\frac{3Fh}{10m}}$

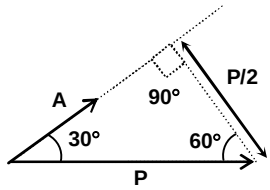


7. A, C, D

Sol. Let linear momentum of the ball A is P.

Minimum possible linear momentum of 'B' is P/2

So linear momentum of 'B' = $\frac{P}{2} \cdot \frac{2}{\sqrt{3}} = \frac{P}{\sqrt{3}}$

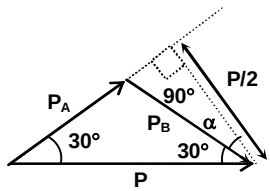


$\cos\alpha = \frac{P/2}{P/\sqrt{3}}$

$\cos\alpha = \frac{\sqrt{3}}{2}$

$\alpha = 30^\circ$

Hence $P_A \cos 30^\circ + P_B \cos 30^\circ = P$



$P_A \frac{\sqrt{3}}{2} + \frac{P}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2} = P$

$P_A \frac{\sqrt{3}}{2} = \frac{P}{2} \Rightarrow P_A = \frac{P}{\sqrt{3}}$

SECTION – B

8. 190

Sol. Let 't' be the time taken by the activity (R) to drop to $\left(\frac{1}{80}\right)$ of its present unsafe value (R_0)

$\frac{R}{R_0} = \frac{1}{80}, \frac{\lambda N}{\lambda N_0} = \frac{1}{80}$

$\Rightarrow \frac{N}{N_0} = \frac{1}{80}$

$$\frac{N_0 2^{-t/t_{1/2}}}{N_0} = \frac{1}{80}$$

$2^{t/30} = 80$

$\Rightarrow t = 190$ days

9. 4552

Sol. For first vessel.

$\Delta U = 0$

$$\Delta Q_1 = \Delta W = nRT\ell n\left(\frac{V_f}{V_i}\right) = 1.R.400.\ell n2$$

$= (0.69) \times 400 \times R = 276\ R$

For second vessel $\Delta V = 0$

$$\Delta Q_2 = \Delta U = nC_V\Delta T = 1\cdot\frac{5}{2}R\Delta T$$

$\Delta Q_1 = \Delta Q_2$

$$276R = \frac{5}{2}R\Delta T$$

$\Rightarrow \Delta T = 110.4$

$T_f = 273 + 127 + 110.4 = 510.4$

$U_{mix} = U_1 + U_2$

$(n_1 + n_2)\, C_{Vmix} T_{mix} = n_1 C_{V_1} T_1 + n_2 C V_2 T_2$

$$T_{mix} = \frac{n_1T_1 + n_2T_2}{n_1 + n_2} = 455.20\,K$$

10. 320

Sol. $$y = \left(\frac{x}{2}\right)^2 = \frac{x^2}{4}$$

$$R = \frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$$

$$y = \frac{x^2}{4}$$

$$\frac{dy}{dx} = \frac{x}{2},\ \left.\frac{dy}{dx}\right|_{x=\frac{3}{2}} = \frac{3}{4}$$

$$\frac{d^2y}{dx^2} = \frac{1}{2}$$

$$R = \frac{\left[1+\left(\frac{3}{4}\right)^2\right]^{3/2}}{1/2} = \frac{125}{32}\,m$$

Force acting on the particle normal to the trajectory $F = \frac{mv^2}{R} = 320$ N

11. 125

Sol. $\frac{1}{2}mv^2 = \mu mgS$

$v = 5$ m/s

Impulse, $F\Delta t = (mv - 0)$

$(i\ell B)\Delta t = 20 \times 5$ …(i)

$i\Delta t = \Delta q$ …(ii)

$\Delta q = \frac{100}{B\ell} = 12.50$ Coulomb

12. 1727

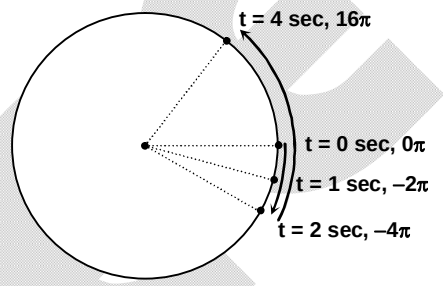
Sol. $\theta = (t^3 - 3t^2)\pi$

$\omega = (3t^2 - 6t)\pi$

When $\omega = 0 \Rightarrow t = 0, t = 2$

$\alpha = (6t - 6)\pi$

Average speed $= \frac{22\pi \times 6}{3} = 44\pi = 138.16$ m/s



13. 425

Sol. $V_L = L\frac{di}{dt} \Rightarrow Ldi = V_L\,dt$

$L\int di = \int V_L dt = \frac{1}{2} \cdot 20 \cdot 10 = 100$

$L(i_f - i_i) = 100$ Amp

$i_f = 5$ amp

for $t \le 4$,

$V_L = \frac{20}{4}t = 5t$

$\int_0^i Ldi = \int V_L dt = 5\int_0^4 tdt$

$L(i - 0) = \frac{5}{2} \times 16 = 40$

$\Rightarrow i = 2$

for $t \ge 4$

$V_L = \frac{10}{3}(10 - t)$

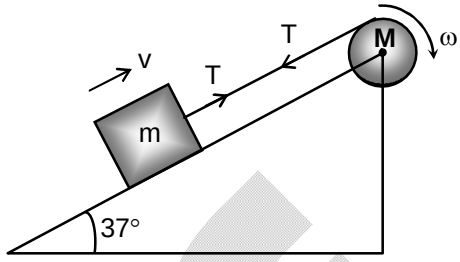
$\int_2^i Ldi = \int_4^7 \frac{10}{3}(10 - t)\,dt$

$\Rightarrow i = 4.25$ amp

SECTION – C

14. 25.00

Sol. $mg\sin37° + \mu mg\cos37° - T = ma$ …(i)



$$TR = \frac{MR^2}{2}\alpha$$

$$T = \frac{M}{2}(R\alpha) = \frac{Ma}{2} \quad \text{…(ii)}$$

From equation (i) and equation (ii)

$$\frac{3}{5}mg + \frac{4}{5}\mu mg = T + \frac{2mT}{M}$$

$$T = \frac{(3+4\mu)mMg}{5(M+2m)}$$

Putting the value T = 25 N

15. 40.00

Sol. Suppose that block moves by a distance 'S' when it comes in rest.

$$\frac{1}{2}\left(\frac{MR^2}{2}\right)(\omega)^2 + \frac{1}{2}mv^2 = mgS\cdot\sin37° + \mu mgS\cdot\cos37°$$

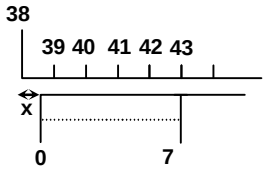
By putting value

S = 40 m

16. 386.25

Sol. X = 5mm – 4.375 mm = 0.625 mm

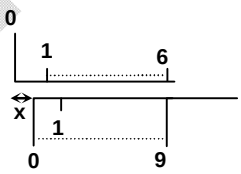
Measured value = 38 mm + 0.625 mm = 38.625 mm



17. 3.75

Sol. $9V = (9)\left(\frac{5}{8}S\right) = 5.625$ mm

X = 6 – 5.625 = 0.375 mm



Chemistry

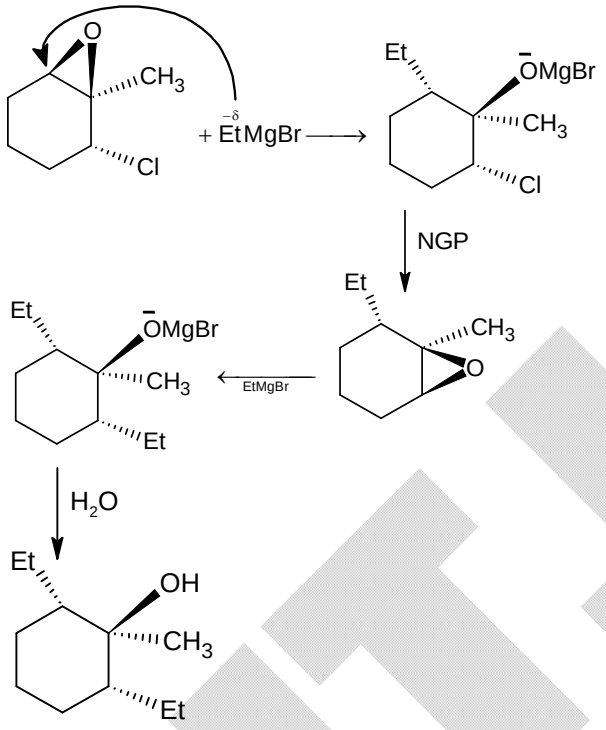
PART – II

SECTION – A

18. D
Sol. The order of pK_a values of ii > iii > i > iv.

19. D
Sol. Due to absence of plane of symmetry. Option (D) is optically active.

20. A
Sol.



21. C
Sol. $KI + O_3 \longrightarrow KOH + I_2 + O_2$

$I_2 + Na_2S_2O_3 \longrightarrow NaI + Na_2S_4O_6$

Law of chemical equivalence,

$[n_e]_{I_2} = (ne)_{Na_2S_2O_3} = (ne)_{O_3}$

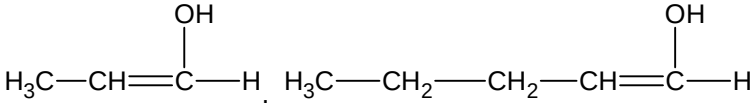
$\therefore [mole]_{O_3} = 1, [mole]_{O_2} = 3$

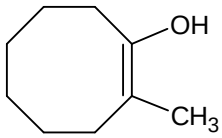
$\therefore$ % Mole % of $O_2 = \frac{3}{4} \times 100 = 75\%$

22. A, C
Sol. High spin complex are $\left[Fe(H_2O)_6\right]^{2+}$ and $\left[Co(H_2O)_6\right]^{2+}$. Low spin complex is $\left[Co(C_2O_4)_3\right]^{3-}$.

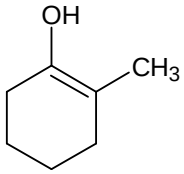
23. B, C
Sol. Polyethene and polypropene are homopolymer. Because they have one type of monomer unit.

24. A, B, D
Sol.





will show the geometrical isomerism



trans form is not possible.

SECTION – B

25. 728

Sol. $(ne)_{charge} = (ne)_{azobenzene} = (ne)_{C_{60}}$

26. 7

Sol. For polytropic process,

PV^x = constant

$$\text{Slope} = -x \times \frac{P}{V} = -x \times \frac{4}{2}$$

$$-1 = -x \times \frac{4}{2} \qquad \therefore x = \frac{2}{4} = 0.5$$

Now,

$$C = \frac{nR}{1-x} + nC_{V,m} \text{ (where C = heat capacity)}$$

$$n = 1, \quad C_{V,m} = \frac{+R}{2} = \frac{3R}{2}$$

$$C = \frac{R}{1-0.5} + 1 \times \frac{3R}{2} = 2R + \frac{3R}{2}$$

$$C = \frac{7R}{2}$$

$$\therefore \text{ Value of } x = \frac{7}{2}$$

$$\therefore 2x = \frac{7}{2} \times 2 = 7.$$

27. 8

Sol. 8 mL 5×10^{-3} [N] $KMnO_4 = 40 \times 10^{-6}$ equivalent of $KMnO_4$.

$$\because (ne)_{CaC_2O_4} = (ne)_{KMnO_4}$$

$$\Rightarrow [\text{moles}] \times n.F = 40 \times 10^{-6}$$

$$\Rightarrow [\text{moles}]_{CaC_2O_4} = \frac{40 \times 10^{-6}}{2} = 20 \times 10^{-6} \text{ mole}$$

$$\therefore [CaC_2O_4] = \frac{20 \times 10^{-6}}{200} \times 1000 = 10^{-4} \text{ mole/L}$$

$$\because CaC_2O_4 \rightleftharpoons Ca^{2+}(aq) + C_2O_4^{2-}$$

$$\therefore K_{SP} = \left[Ca^{2+}\right]\left[C_2O_4^{2-}\right] = S^2 = \left[10^{-4}\right]^2 = 10^{-8}$$

$$\therefore \text{Value of } -\log K_{SP} = -\log 10^{-8} = 8$$

28. 14

Sol. $K_2Cr_2O_7 + 14HCl \longrightarrow 2KCl + 3Cl_2 + 2CrCl_3 + 7H_2O$

29. 24

Sol. Number of stereo isomers = 8

Number of enantiomeric pair = 4

$\therefore$ Number of diastereomeric pair $8_{C_2} - 4 = 24$

30. 4

Sol. Statement (a) (b) (d) and (e) are correct and statement (b) is wrong.

SECTION – C

31. 1.60

Sol. Let x mole, NH_4Cl was initially present.

$NH_4Cl(s) \longrightarrow NH_3(g) + HCl(g)$

x x

$PV = nRT \Rightarrow 114 \times V = 0.01 \times R \times 300$ …(1)

and, $908 \times V = (0.01 + 2x) \times R \times 600$ … (2)

$\therefore x = 0.015$

$\therefore [Mass]_{NH_4Cl} = x \times 53.5 = 0.8$ gram

$\therefore$ Value of 2y is 1.6 gram.

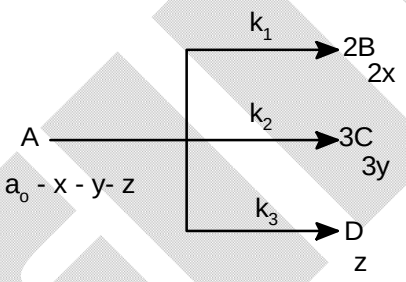
32. 1.64

Sol. $V = \dfrac{0.01 \times 0.0821 \times 300 \times 760}{114} = 1.642$ L

33. 0.36

34. 1.08

Sol. **(for the Q. No. 33 and 34.)**



$A_t = a_o x^{-(k_1+k_2+k_3)t} = 1 \times e^{-0.5 \times 4.606}$

$A_t = \dfrac{1}{10} = 0.1$

$\therefore \dfrac{k_1}{k_2} = \dfrac{x}{y} \Rightarrow \dfrac{1}{2} = \dfrac{x}{y}$ … (1)

$\dfrac{k_2}{k_3} = \dfrac{y}{z} \Rightarrow y = z$ … (2)

$\therefore x + y + z = 0.9$ … (3)

$B_t = 2x = 0.36$ M, $C_t = 3y = 1.08$ M

$D_t = z = 0.36$ M

Mathematics

PART – III

SECTION – A

35. B

Sol. $\Delta = \begin{vmatrix} 1+1+1 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix}^2$

$= \begin{vmatrix} 1 & 0 & 0 \\ 1 & \alpha-1 & \beta-1 \\ 1 & \alpha^2-1 & \beta^2-1 \end{vmatrix}^2$ [$C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$]

$= [(\alpha-1)(\beta^2-1) - (\alpha^2-1)(\beta-1)]^2 = [\alpha\beta(\beta-\alpha) + (\beta-\alpha) + (\alpha^2-\beta^2)]^2$

$= [\alpha\beta + 1 - (\alpha+\beta)]^2\ [(\alpha+\beta)^2 - 4\alpha\beta]$

$= \left(\frac{c}{a}+\frac{b}{a}+1\right)^2\left(\frac{b^2}{a^2}-\frac{4c}{a}\right) = \frac{(a+b+c)^2(b^2-4ac)}{a^4} > 0$ if roots are real and distinct

36. A

Sol. We have, $\sum_{r=0}^{n}\sum_{s=0}^{n} C_r C_s = \left(\sum_{r=0}^{n} C_r^2\right) + 2\sum_{0\le r<s\le n}\sum C_r C_s$

$\Rightarrow 2^{2n} = {}^{2n}C_n + 2\sum_{0\le r<s\le n}\sum C_r C_s$

$\Rightarrow \sum_{0\le r<s\le n}\sum C_r C_s = \frac{1}{2}\left[2^{2n} - {}^{2n}C_n\right]$

37. A

Sol. $\lim_{h\to 0}\frac{f(2+h)}{\sin h} = 3$

For finite limit, f(2) = 0

$\therefore \lim_{h\to 0}\frac{f(2+h)-f(2)}{\left(\frac{\sin h}{h}\right)\times h} = f'(2) \Rightarrow f'(2) = 3$

38. C

Sol. f(f(1)) = 0 $\Rightarrow$ f(1 + a + b) = 0

f(f(2)) = 0 $\Rightarrow$ f(4 + 2a + b) = 0

$\therefore$ 1 + a + b and 4 + 2a + b are roots of f(x)

$\therefore$ 3a + 2b + 5 = –a

$\Rightarrow$ 4a + 2b + 5 = 0 ..... (1)

(a + b + 1)(2a + b + 4) = b

$\Rightarrow (a+b+1)\left(-\frac{5}{2}+4\right) = b \Rightarrow (a+b+1)\left(\frac{3}{2}\right) = b$

$\Rightarrow$ 3a + 3b + 3 = 2b

$\therefore$ f(0) = b

$\Rightarrow$ b = –3a – 3 ..... (2)

From equation (1), we get

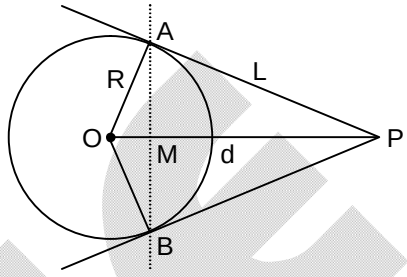
4a + 2(–3a – 3) + 5 = 0

$\Rightarrow -2a - 1 = 0 \Rightarrow a = -\frac{1}{2}$

$\therefore\ b = -3\left(-\frac{1}{2}\right) - 3 = -\frac{3}{2},\ f(0) = b = -\frac{3}{2}$

39. A, C

Sol. $\frac{1}{(OA)^2} + \frac{1}{(PA)^2} = \frac{1}{16}$



$\frac{1}{R^2} + \frac{1}{L^2} = \frac{1}{16} \Rightarrow \frac{L^2 + R^2}{L^2 \cdot R^2} = \frac{1}{16}$

$\Rightarrow \frac{d^2}{L^2R^2} = \frac{1}{16} \Rightarrow L.R = 4d$

$\therefore$ Area of quadrilateral OAPB = RL = 4(OP)

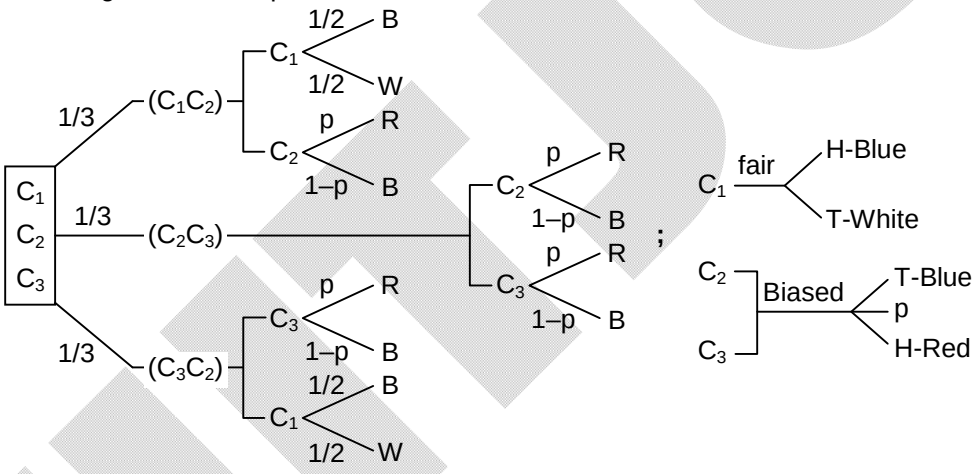
Again, area of ΔOAP

$\frac{1}{2}RL = \frac{1}{2}d \cdot AM \Rightarrow 4d = d \cdot AM \Rightarrow AM = 4$

$\therefore$ AB = 8 units

40. C, D

Sol. Tree diagram of the experiment is as shown:



P (both coins show up the same colour)

$= \frac{1}{3} \cdot \frac{1}{2}(1-p) + \frac{1}{3}\left[p^2 + (1-p)^2\right] + \frac{1}{3} \cdot \frac{1}{2}(1-p) = \frac{1}{3}(1-p) + \frac{1}{3}\left(2p^2 - 2p + 1\right) = \frac{1}{3}\left(2p^2 - 3p + 2\right)$

$\frac{29}{96} = \frac{1}{3}\left[2p^2 - 3p + 2\right]$ (Given)

$\frac{29}{96} = 2p^2 - 3p + 2 \Rightarrow 29 = 64p^2 - 96p + 64$

$\therefore\ 64p^2 - 96p + 35 = 0$ or $64p^2 - 56p - 40p + 35 = 0$

$8p(8p - 7) - 5(8p - 7) = 0 \Rightarrow (8p - 7)(8p - 5) = 0\ \therefore\ p \cdot \frac{7}{8}$ or $\frac{5}{8}$

41. A, C, D

Sol. Each one should get $(4\lambda + 2)\ \lambda \in w$ blankets

$(4\lambda_1 + 2) + (4\lambda_2 + 2) + (4\lambda_3 + 2) + (4\lambda_4 + 2) = 32$

$4(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4) + 8 = 32 \Rightarrow \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = \dfrac{24}{4} = 6$

Number of ways = ${}^{6+4-1}C_{4-1} = {}^{9}C_3$

SECTION – B

42. 8

Sol. Let $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Now, $M\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \Rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \Rightarrow \begin{bmatrix} a-b \\ c-d \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

So, $a - b = -1$ ..... (1)

$c - d = 2$ ..... (2)

Also, $M^2\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow M\left(M\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow M\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -a+2b \\ -c+2d \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

So, $-a + 2b = 1$ ..... (3)

$-c + 2d = 0$ ..... (4)

$\therefore$ From (1), (2), (3), (4), we get $a = -1, b = 0, c = 4, d = 2$

Hence, $M = \begin{bmatrix} -1 & 0 \\ 4 & 2 \end{bmatrix}$

Now, $\det(M - xI) = 0$ (Given)

$\Rightarrow \begin{bmatrix} -1 & 0 \\ 4 & 2 \end{bmatrix} - x\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} -1-x & 0 \\ 4 & 2-x \end{bmatrix} = 0$

$\Rightarrow -(1 + x)(2 - x) = 0 ; (x + 1)(x - 2) = 0$

$x = 2, -1 ; x_1 = 2, x_2 = -1$

$5x_1 + 2x_2 = 10 - 2 = 8$

43. 6

Sol. $SS_1 = T^2$

$(y^2 - 4x)\left(y_1^2 - 4x_1\right) = \left(yy_1 - 2\left(x + x_1\right)\right)^2$

$(y^2 - 4x)(4 + 4) = [2y - 2(x - 1)]^2 = 4(y - x + 1)^2$

$2(y^2 - 4x) = (y - x + 1)^2$

Solving with the line $x = 2$, we get

$2(y^2 - 8) = (y - 1)^2$ or $2(y^2 - 8) = y^2 - 2y + 1$

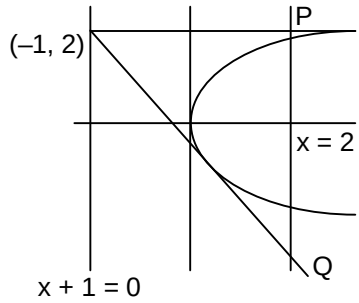
Or $y^2 + 2y - 17 = 0$

Where $y_1 + y_2 = -2$ and $y_1y_2 = -17$

Now $|y_1 - y_2|^2 = (y_1 + y_2)^2 - 4y_1y_2$

Or $|y_1 - y_2|^2 = 4 - 4(-17) = 72$

$\therefore \left(y_1 - y_2\right) = \sqrt{72} = 6\sqrt{2}$



44. 376

Sol. We have $\tan^{-1}\left(x + \dfrac{3}{x}\right) - \tan^{-1}\left(x - \dfrac{3}{x}\right) = \tan^{-1}\dfrac{6}{x}$

$\Rightarrow \tan^{-1}\left(\dfrac{\left(x + \frac{3}{x}\right) - \left(x - \frac{3}{x}\right)}{1 + \left(x + \frac{3}{x}\right)\left(x - \frac{3}{x}\right)}\right) = \tan^{-1}\dfrac{6}{x} \Rightarrow x^2 - \dfrac{9}{x^2} = 0 \Rightarrow x^4 = 9$

Hence, $(5x^8 - 4x^4 + 7) = 5(81) - 4(9) + 7 = 405 - 36 + 7 = 412 - 36 = 376$

45. 51

Sol. $[1\ \ 25]\begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}^5 \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}^{10} \begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}^5 \begin{bmatrix} x^2-5x+20 \\ x+2 \end{bmatrix} = [40]$

Let $A = \begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$

Here, AB = BA = I

$\therefore A^5 B^{10} A^5 = I$

$[1\ \ 25]\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x^2-5x+20 \\ x+2 \end{bmatrix} = [40] \Rightarrow [1\ \ 25]\begin{bmatrix} x^2-5x+20 \\ x+2 \end{bmatrix} = [40]$

$x^2 - 5x + 20 + 25x + 50 = 40 \Rightarrow x^2 + 20x + 30 = 0$

$(1-\alpha)(1-\beta) = 1 - (\alpha+\beta) + \alpha\beta = 1 - (-20) + 30 = 51$

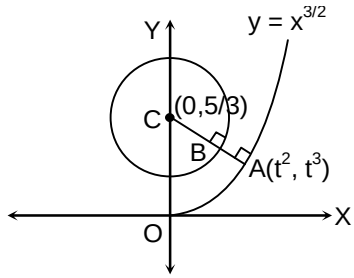
46. 2

Sol. We have $9x^2 + 9y^2 - 30y + 16 = 0$

Which is a circle having centre $\equiv \left(0, \frac{5}{3}\right)$ and radius = $\sqrt{\left(\frac{5}{3}\right)^2 - \frac{16}{9}} = 1$

Let us choose any point on the curve $y^2 = x^3$ as $A(t^2, t^3)$

If B is the point on the circle and nearest to A, then



AB = AC – radius = $\sqrt{(t^2-0)^2 + \left(t^3 - \frac{5}{3}\right)^2} - 1$

Let $f(t) = t^4 + \left(t^3 - \frac{5}{3}\right)^2$

And $f'(t) = 4t^3 + 2\left(t^3 - \frac{5}{3}\right)\cdot 3t^2 = 2t^2\left[2t + 3\left(t^3 - \frac{5}{3}\right)\right]$

$= 2t^2(3t^3 + 2t - 5) = 2t^2(t-1)(3t^2 + 3t + 5)$

The value of 't' at which AB attains minima is given by the equation $f'(t) = 0$ gives $t = 0, 1$

But $t = 0$ is not a point of extrema, since $f'(t)$ does not change sign in the neighbourhood of $t = 0$.

However, $t = 1$ is a point of minima, since $f'(1-) < 0$ and $f'(1^+) > 0$. Hence, the minimum value of AB is

$\sqrt{1 + \left(1 - \frac{5}{3}\right)^2} - 1 \simeq 0.2$

47. 5

Sol. Let the equation of line on which these three points lie be $lx + my + n = 0$ and the point $\left(\frac{t^3}{t-1}, \frac{t^2-3}{t-1}\right)$ lie on the line where t = a, b, c $l\left(\frac{t^3}{t-1}\right) + m\left(\frac{t^2-3}{t-1}\right) + n = 0$

$lt^3 + m(t^2 - 3) + n(t - 1) = 0 \Rightarrow t^3 l + t^2 m + tn - 3m - n = 0$

If a, b, c are the roots of given equation, then

$a + b + c = \frac{-m}{l}$ ..... (1)

$ab + bc + ca = \frac{n}{l}$ ..... (2)

$abc = \frac{3m}{l} + \frac{n}{l}$ ..... (3)

Using (1), (2) and (3), we get $abc = -3(a + b + c) + ab + bc + ca$

$abc + 3(a + b + c) = ab + bc + ca \Rightarrow \alpha + \beta + \gamma = 1 + 3 + 1 = 5$

SECTION – C

48. 1.00

49. 1.00

Sol. (for Q. 48 and 49)

$\because \int_0^{2x} t^2 f(2x-t)\,dt = 2e^{2x} - 4x^2 + 2x - 2$

Using king, we get

$\Rightarrow \int_0^{2x} (2x-t)^2 f(2x-(2x-t))\,dt = 2e^{2x} - 4x^2 + 2x - 2$

$\Rightarrow 4x^2 \cdot \int_0^{2x} f(t)\,dt - 4x \cdot \int_0^{2x} t f(t)\,dt + \int_0^{2x} t^2 f(t)\,dt = 2e^{2x} - 4x^2 + 2x - 2$

Differentiation with respect to x

$\Rightarrow 4x^2 f(2x)2 + 8x \cdot \int_0^{2x} f(t)\,dt - 8x^2 f(2x)2 - 4 \cdot \int_0^{2x} t f(t)\,dt + 4x^2 f(2x) \cdot 2 = 4e^{2x} - 8x + 2$

$\Rightarrow 8x \cdot \int_0^{2x} f(t)\,dt - 4 \cdot \int_0^{2x} t f(t)\,dt = 4e^{2x} - 8x + 2$

Again differentiating, we get $8xf(2x) \cdot 2 + 8\int_0^{2x} f(t)\,dt - 8xf(2x) \cdot 2 = 8e^{2x} - 8$

$\Rightarrow \int_0^{2x} f(t)\,dt = e^{2x} - 1 \Rightarrow f(2x) \cdot 2 = 2e^{2x} \Rightarrow f(2x) = e^{2x}$

$\therefore f(x) = e^x \therefore f^{-1}(x) = \ln x \therefore f^{-1}(e) = 1$

49. $\because g(x) + g'(x) = e^{-x} \Rightarrow \int e^x (g(x) + g'(x))\,dx = \int 1 \cdot dx$

$\Rightarrow e^x g(x) = x + c \because g(1) = \frac{1}{e} \therefore c = 0 \Rightarrow g(x) = xe^{-x}$

$\therefore$ Area bounded $= \int_0^{\infty} xe^{-x}dx = \left(-xe^{-x}\right)_0^{\infty} + \int_0^{\infty} e^{-x}dx = 0 - \left(e^{-x}\right)_0^{\infty} = -(0-1) = 1$

50. 2.00

51. 1.00

Sol. (for Q. 50 and 51)

$A(t) = \frac{(a-b)(a-t)(b-t)}{2abt}$

$A'(t) = \frac{(a-b)(t^2-ab)}{2abt^2}$

So, $t = \sqrt{ab}$ is point of maxima moreover

$\int_a^{\sqrt{ab}} \left(-\frac{1}{at}x + \frac{1}{a} + \frac{1}{t} - \frac{1}{x}\right)dx = \int_{\sqrt{ab}}^{b} \left(-\frac{1}{bt}x + \frac{1}{b} + \frac{1}{t} - \frac{1}{x}\right)dx$

